Silva Nurzanatul Dahmalena

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ABOUT ME

Informatics Engineering student with a strong passion for Data Analytics and Business Intelligence. Experienced in analyzing business data using Python, SQL, and Microsoft Excel to uncover patterns and generate actionable insights. Skilled in building interactive dashboards with Streamlit and supporting ETL processes to ensure data quality. Proficient in statistical analysis and data visualization, with a collaborative mindset and strong communication skills to translate data into business recommendations.

EDUCATION

STT Terpadu Nurul Fikri **2024 - Present (Expected 2028)**

- Bachelor of Information Technology
- GPA: 3.90/4.00
- Recipient of KIP-K Scholarship (National Merit Scholarship)

SKILL

- **Technical Skills:** Python, Pandas, NumPy, Scikit-Learn, SQL (MySQL), Machine Learning, Natural Language Processing (NLP), FAISS, Gradio, Streamlit, Microsoft Excel, KNIME, Git/GitHub, Data Visualization, Google Colab
- **Soft Skills:** Analytical Thinking, Problem Solving, Teamwork, Adaptability, Communication

PROJECT EXPERIENCE

Traffic Volume Prediction for Smart Mobility in Bogor City **April - Juli 2026**

- Analyzed 10+ years of traffic volume data (2015-2025) using Python, Pandas, and NumPy for data cleaning, EDA, and feature engineering.
- Developed and compared Linear Regression and Random Forest models to forecast traffic volume patterns.
- Evaluated model performance using R^2, MAE, and RMSE metrics, identifying Random Forest as the best-performing model (R^2: 0.89, MAE: 12.3).
- Built an interactive Streamlit dashboard to visualize traffic trends, feature importance, and real-time prediction results.
- Collaborated in a team of 3 to design the dashboard layout and integrate predictive models into the application.

Boarding House Database Management System | Project Leader **December 2024 - January 2025**

- Led a team of 3 members in designing and implementing a complete database management system for boarding house operations using MySQL Workbench.
- Designed Entity Relationship Diagram (ERD) consisting of 5 integrated tables to manage owner, tenant, room, facility, and payment data.
- Developed SQL Triggers and Stored Procedures to automate room status updates and payment transactions, reducing manual data entry time by approximately 40%.
- Implemented Data Control Language (DCL) to manage user access permissions (GRANT/REVOKE), ensuring data security and role-based access control.
- Documented the entire database schema and created user guides for system maintenance and future development.

Analysis of Income, Urbanization, and Internet Usage Patterns Across Countries **Juni - Juli 2025**

- Conducted Exploratory Data Analysis (EDA) on the Gapminder Internet dataset to analyze relationships between income per capita, urbanization rate, and internet usage across multiple countries.
- Utilized Python (Pandas, NumPy), KNIME, and Microsoft Excel for data cleaning, descriptive statistical analysis, and data visualization.
- Applied Pearson and Spearman correlation analysis to identify the strength of relationships between variables.
- Developed simple and multiple linear regression models to measure the impact of income per capita and urbanization rate on internet usage.
- Generated key business insight: urbanization has a stronger influence on internet usage than income per capita → indicating that infrastructure development may be more critical than economic growth for digital adoption.

CERTIFICATIONS

- **Intro to Data Analytics | RevoU | 2026**
- **SQL (Basic) | HackerRank | 2026**
- **Google Analytics Certification | Google Skillshop | 2026**
- **Get Started Using Google Analytics | Google Skillshop | 2026**

ORGANIZATION EXPERIENCE

GWD (Growth With Data) - Public Relations Staff **2026 - Present**

- Managed external communications and promoted data literacy events across campus.

Student Executive Board (BEM) - Advocacy and Student Services Staff **2025 - 2026**

- Planned and executed student development programs, including workshops and seminars (e.g., Mental Health Seminar) to support student well-being and growth.
- Collected and processed student aspirations to inform program planning and policy recommendations.
- Coordinated with multiple divisions to ensure successful event implementation.

Student Representative Council (DPM) | Commission II (Student Aspirations) Staff **2026 - Present**

- Collected and processed student aspiration data to support policy recommendations.
- Coordinated and facilitated discussions with other student organizations (BEM, HIMA) to find collaborative solutions for student issues.
- Advocated for student interests by presenting aspiration data and proposed solutions to campus administration.

COMMITTEE EXPERIENCE

SeminarMental Health **2025**

- Assisted in the preparation and execution of the seminar to ensure smooth event operations.
- Coordinated with committee members to support event logistics and participant needs.

Debat Rakyat Kampus (DERAS) **2025**

- Managed the procurement and distribution of meals and refreshments for committee members and participants.
- Ensured the availability of food and beverages throughout the event.

International Campus Talk – Secretary **2025**

- Managed event administration and documented meeting minutes.
- Assisted in coordinating communication and activities among committee members.
- Maintained event documentation and supported organizational planning.

Department of Informatics Orientation Program – Event Staff **2025**

- Contributed to planning and coordinating an orientation program attended by 150+ new students.
- Managed event schedules and ensured activities were conducted according to the planned timeline.
- Collaborated with multiple divisions to support operational and technical requirements.
- Facilitated ice-breaking sessions to encourage participant engagement and interaction.

Student Discussion Forum (SIGMA) – Master of Ceremony (MC) **2025**

- Hosted and moderated a panel discussion involving three campus representatives.
- Prepared and improvised opening, transition, and closing scripts to maintain audience engagement.
- Collaborated with the organizing committee to ensure smooth session transitions and adherence to the event schedule.